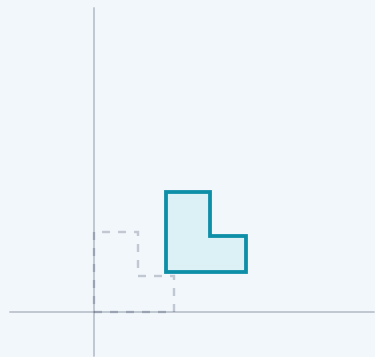


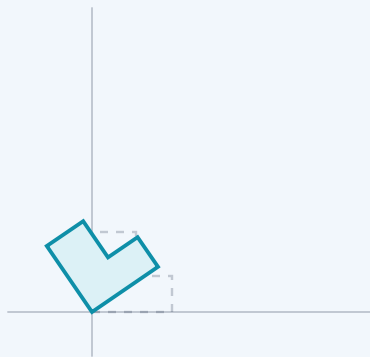
Translation



$$y = x + b$$

add a vector

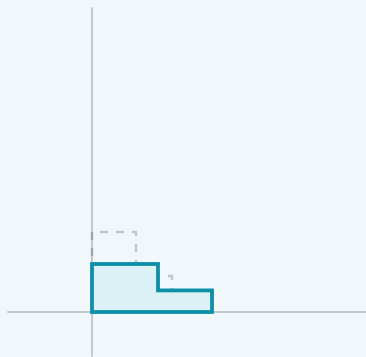
Rotation



$$y = R \cdot x$$

multiply by a rotation matrix

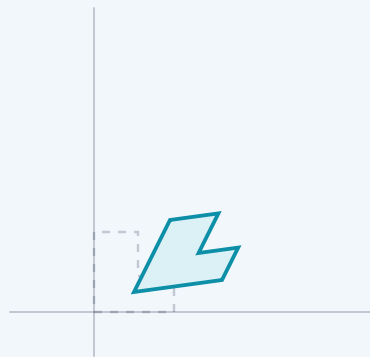
Scaling



$$y = D \cdot x$$

multiply by a diagonal matrix

Affine



$$y = W \cdot x + b$$

any matrix, then a vector

Dashed: before. Solid: after. Every layer in a network does the last one — a matrix, then a vector.